The EAGLET-2 L1D Processor: Operational Cloud Masks and Natural Colour in the IRIDE Ground Segment

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EAGLET-2 is an IRIDE optical mission based on small satellites that carry an RGB camera.

Per-pixel cloud and cloud-shadow maps for every product, despite the lack of cirrus and thermal bands and of labelled training data: OmniCloudMask [3], a sensor-agnostic model, runs as published.

A dedicated L1D product type is introduced within the IRIDE ground segment, carrying these maps and a natural-colour image.

1 EAGLET-2 in the IRIDE ground segment

IRIDE is the Italian Earth observation "constellation of constellations"; its development is coordinated by ESA with the support of ASI [1, 2]. EAGLET-2, built by OHB Italia, is one of its optical constellations [2]. Its Payload Data Ground Segment (PDGS) takes EAGLET-2 data from raw to L1D through five processors, the L1D processor being the last.

L0 processor — Reference orbit scenario file
→ **L0 product** · raw data

L1A processor — Non-uniformity correction matrices; Bad pixel replacement matrix
→ **L1A product** · raw counts corrected for detector non-uniformity, bad pixels replaced

L1B processor — Absolute calibration coefficients; Geometric model; DEM
→ **L1B product** · top-of-atmosphere radiance

L1C processor — GCP; DEM; Spectral solar irradiance
→ **L1C product** · top-of-atmosphere reflectance, orthorectified

L1D processor — RGB LUT (colour AUX file); this work
→ **L1D product** · cloud mask, cloud confidence and natural colour

Each processor also reads its processing parameters; GCP and DEM are read-only volumes.

2 Running the L1D processor in the PDGS

PDGS processing facility

1. A new L1C (1CSP_RGB3) reaches the PDGS archive.
2. The Task Table asks for one L1C, the colour AUX file if any, and 2 CPUs.
3. The PDGS prepares a working directory with the L1C, the AUX file and a Job Order.
4. It starts the container, with the Job Order path as its only argument.

L1D processor, a Docker image

Strict and self-contained: anything unexpected ends with exit code 255 and no product. The model weights travel inside the image, nothing goes over the network, and a scene of about 20 × 16 km takes 3 to 5 minutes on 2 CPUs.

5. **Read** the Job Order and the L1C (RGB, 16-bit DN); work offline, logging on stdout and stderr.
6. **Calibrate** to top-of-atmosphere reflectance, DN × gain + offset.
7. **Prepare** the valid-data footprint; resample the RGB from about 2 m to 5 m.
8. **Segment** with OmniCloudMask: four class probabilities per pixel, on CPU.
9. **Classify** at full resolution: cloud where COM = P(thick) + P(thin) is 0.5 or more, thick or thin by the larger; shadow where P(shadow) > P(clear).
10. **Correct** the colour with the RGB LUT from the AUX file (section 5).
11. **Package** the L1D (natural-colour COG, CLM, COM, previews with a KML, STAC and XML) and exit: 0 when done, 255 when it fails.

12. The PDGS collects the L1D for the archive and removes the working directory.

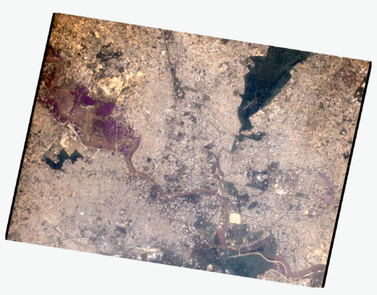
3 Clear scenes stay clear



Atlanta, USA
IME02 · 17 Oct · 0.4 % cloud



New Orleans, USA
IME03 · 17 Oct · 0.0 % cloud



Lucknow, India
IME03 · 18 Oct · 0.0 % cloud



Bakshi Ka Talab, India
IME01 · 18 Oct · 0.0 % cloud

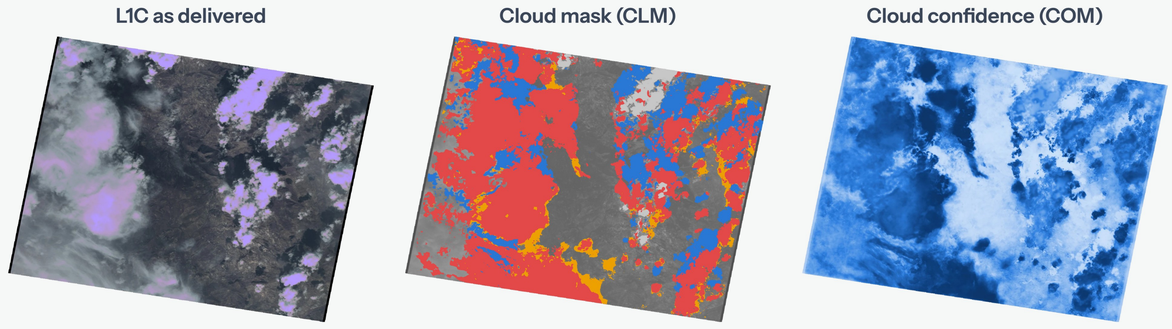


New York, USA
IME03 · 19 Oct · 0.0 % cloud

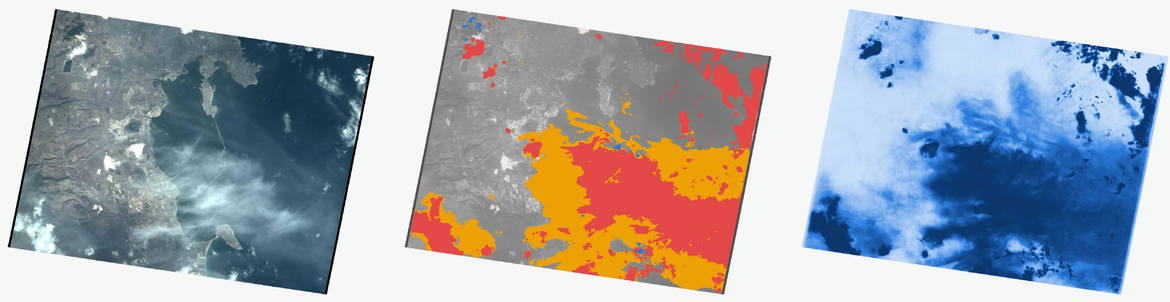
Five of the seven test scenes, from three satellites, are cloud-free. The processor reports at most 0.4 % cloud on them, in the STAC and XML of each L1D.

4 Cloud and shadow masks on EAGLET-2

OmniCloudMask runs on the EAGLET-2 RGB alone. In the cloud mask (CLM) the image is shown in grey with the classes on top; the cloud confidence (COM) runs from 0 to 1.



Burgio, Sicily · IME08 · 17 Oct 2026 · cumulus and their shadows; cloud 43.9 % (thick 39.5, thin 4.4), shadow 13.3 %



Syracuse, Sicily · IME04 · 17 Oct 2026 · haze over the sea and small cumulus; cloud 43.5 % (thick 21.5, thin 22.0), shadow 0.2 %

clear (the image in grey) · thick cloud · thin cloud · cloud shadow

COM: 0 — 0.5, the CLM cloud line — 1

5 Natural colour for the L1D

Due to a near-infrared tail in the spectral response of the red band, vegetation appears purple in the L1C RGB composite. The L1D processor mitigates this effect with a LUT derived from a comparison with Sentinel-2.

1 Match with Sentinel-2
Same ground in both, at most 3 days apart; cloud-free pixels aligned at 10 m

2 Measure the leak
How much near-infrared each EAGLET-2 band picks up: the red band the most

3 Build the LUT
Per colour cell and vegetation around: factor = Sentinel-2 ÷ EAGLET-2, band by band

4 Correct each pixel
Look up the pixel's colour cell, then, band by band, L1D = L1C × factor



L1C as delivered

One forest pixel, step by step

	Red	Green	Blue	NIR
Sentinel-2 sees	0.045	0.070	0.085	0.30
EAGLET-2 sees	0.138	0.110	0.136	
LUT factor	**× 0.32**	**× 0.64**	**× 0.62**	
L1D	0.044	0.071	0.085	

Typical top-of-atmosphere reflectance. The EAGLET-2 values are modelled with the estimated leak; the factors are read from the AUX table, for a pixel among vegetation.



L1D



Sentinel-2

New Orleans (IME03): the same 2 km at 10 m, with the same display stretch for all three. **The purple is gone;** a slight blue cast remains in the L1D, the next target for the LUT.

References

[1] G. Costa, T. Rumpf, E. G. Cadau, L. Sapia, V. Faccin, A. Papa, and C. Stella, "IRIDE: The New Italian Space Programme," in Proc. IEEE IGARSS, 2024, pp. 6663–6665.

[2] E. Cadau, A. Minchella, V. Faccin, L. Festa, L. Marques, A. Papa, G. Costa, and C. Stella, "The IRIDE Project: Satellites, Algorithms and Products of the New Italian Space Programme," in Proc. IEEE IGARSS, 2024, pp. 6763–6766.

[3] N. Wright, J. M. A. Duncan, J. N. Callow, S. E. Thompson, and R. J. George, "Training sensor-agnostic deep learning models for remote sensing: Achieving state-of-the-art cloud and cloud shadow identification with OmniCloudMask," Remote Sensing of Environment, 2025.

Contains modified Copernicus Sentinel data (2026).

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